

Eigenvalue Analysis of Close Coupled 13.56 MHz RFID-Labels

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Abstract — The technology of RFID (Radio Frequency Identification) is on the way to change industrial processes but also our daily life significantly, as a variety of applications based on this technology are on the way to be implemented. While typical applications in the past had to handle only one tag/label in the field, future applications will have to face many labels in the field. In particular if they show a high coupling factor between them a severe detuning will occur, leading to a loss of functionality in the worst case. Examples for such scenarios may be smart shelves where goods are equipped with RFID tags or identification cards, passports etc. carried close together. It is topic of this work to characterize the physical behavior of such close coupled systems in terms of effected resonance frequency and others. The influence will be characterized based on mathematical analysis - in particular an eigenvalue analysis. It will be pointed out that a solid mathematical understanding and modeling is absolutely necessary to handle future applications and to guarantee their functionality.

Index Terms — 13.56 MHz RFID Systems, close coupled systems, eigenvalue decomposition

I. INTRODUCTION

Applications based on Radio Frequency Identification (RFID) first appeared in the 1980s and since then this technology started to influence our daily life significantly. While the original applications and intentions were tracking and access applications, this technology is being used in a wide range of applications nowadays and will be used in a significant broader area of applications in future [1].

Different aspects are responsible for this rapidly growing market. While it is in particular the improvement of the IC technology (low power, low price, low size) on the one hand, it is a significant better physical understanding and modeling of RFID systems on the other hand - so for example the improved techniques for multiple tag reading, as it will be the case for several tags in the field.

The underlying investigations are dedicated to passive systems, as when talking of high volume and therefore low cost applications almost only passive systems can be taken into account. This means that the microcircuit has to power itself and provide power for a reply signal from the amount of energy that it can extracted from an interrogation (reader) signal.

Additionally it is to mention that our investigations are based on the frequency of 13.56 MHz although there is a strong

tendency towards UHF (>800 MHz). Nevertheless, it is to say that for a variety of applications HF may be the better choice than UHF [2].

II. BASIC FUNCTIONALITY OF PASSIV 13.56 MHz SYSTEMS

The function of a passive Radio Frequency Identification System can be described in its simplest form as that of an RFID reader identifying an RFID tag, reading data from the tag and writing data to the tag - without contact or line of sight. In the 13 MHz area we are (primarily) talking of inductively coupled systems where the necessary energy is provided by the magnetic field of the reader. Figure 1 depicts the very principle of an inductive coupled RFID system, which can be summarized as follows:

For inductive coupled systems the underlying antennas are represented by coils of a defined size. It is well known that a coupling system of two coils can be replaced equivalent by a transformer. The connection between these two coils is given by the magnetic field (B) and the underlying value to describe this connection is the mutual inductance (M) and/or the coupling factor (k). The mentioned physical values are given by the following elementary laws/equations [3]-[6]:

The law of Biot and Savart is given by

$$\vec{B} = \frac{\mu_0 \cdot i_1}{4\pi} \cdot \oint_S \frac{d\vec{s} \times \vec{x}}{|\vec{x}|^3} \quad (1)$$

and allows the calculation of the magnetic field at every point as function of the current i_1 , as well as of geometry. In equation (1) μ_0 describes the permeability, x stands for the distance and s describes the integration-path along the coil. Besides this the mutual inductance and the coupling factor, which is in general easier to handle, are given by

$$M = \int_{A_2} \frac{B(i_1)}{i_1} \cdot dA_2 \quad (2)$$

$$k = \frac{M}{\sqrt{L_1 \cdot L_2}} \quad (3)$$

Here A_2 describes the area of the second coil, while L_1 and L_2 describe the inductance of the two coils.

Figure 1 depicts additionally that due the connection between first and second coil a change of the impedance on transponder side leads to an unequivocal change on the reader

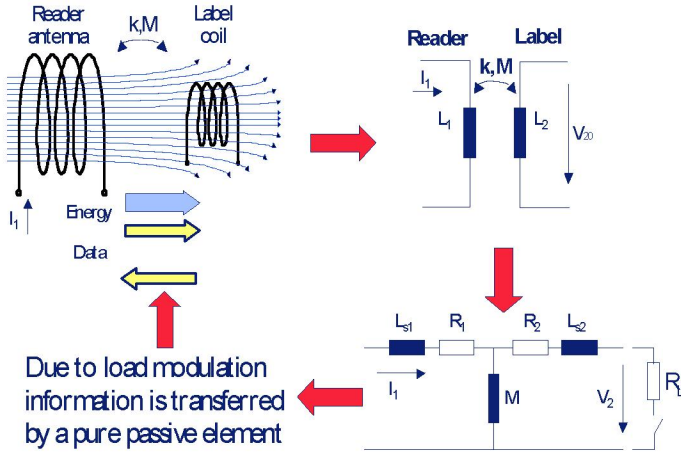


Fig. 1: Physical principal of passive 13.56 MHz Systems.

side (change of the seen impedance). This variation of the impedance on the secondary side, which is realized on the primary side is called load modulation - simply by loading the field more or less, information is transferred.

III. MODELING AND ANALYSIS

Topic of this chapter is a correct modeling and mathematical analysis of multiple labels in the field. Figure 2 points out a model of a typical (resonant) inlet. L_1 describes the inductance of the coil (antenna), R_{L1} describes a parasitic impedance and R_{C1} describes the impedance of the IC which is attached to the antenna. Finally to mention is the capacity C_1 , which includes a parasitic capacity of the coil itself as well as the capacity of the IC - together they build a resonant system with the inductance L_1 tuned to 13.56 MHz. To operate a label close to the resonance frequency is of relevance as only there the necessary voltage is induced and a powering of the IC is possible. Therefore it is of significance to characterize the behavior due to several labels close by precise as a significant

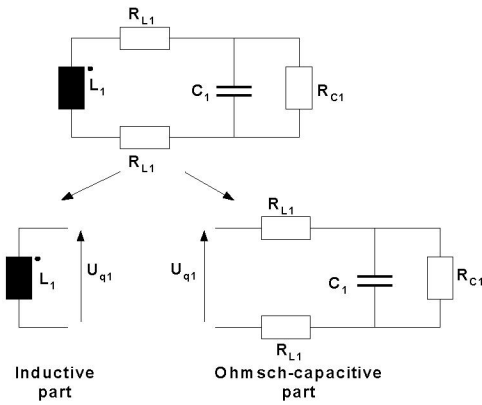


Figure 2: Model of an inlet.

detuning effect is to be expected, preventing an optimal

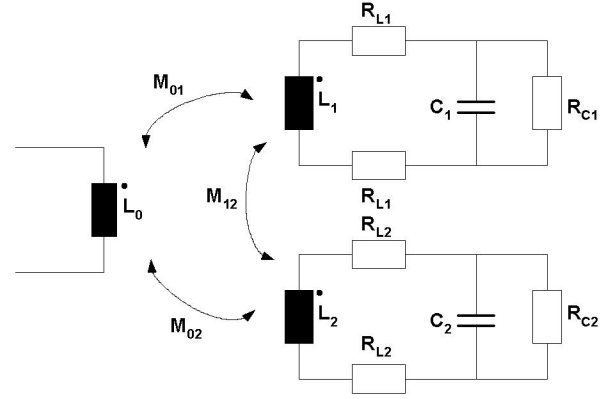


Fig. 3: Coupling System of two labels and one driver coil.

functionality. For the underlying investigations the model of the inlet is separated into an inductive part, containing L and an ohmic-capacitive part, containing R and C .

Taking into account that the influence of several labels will be investigated, figure 3 shows this scenario, pointing out also the mutual inductance (M) between tags. Note that L_0 in figure 3 stands for the principal reader functionality (here a driver coil), which shows an overall flat transfer-function for our investigations.

Based on the previous description, the following mathematical system based on the Ohm's Law (for two and more labels) can be defined for the inductive part:

$$U = A * I + B * i_0, \quad (4)$$

where

$$U = \begin{bmatrix} U_{q1} \\ \vdots \\ U_{qn} \end{bmatrix}, B = -j\omega \begin{bmatrix} M_{01} \\ \vdots \\ M_{0n} \end{bmatrix}, I = \begin{bmatrix} i_1 \\ \vdots \\ i_n \end{bmatrix} \quad (5)$$

$$A = j\omega \begin{bmatrix} L_1 & \dots & M_{1n} \\ \vdots & L_i & \vdots \\ M_{n1} & \dots & L_n \end{bmatrix} \quad M_{1n} = M_{n1} \quad (6)$$

In a first step it is in particular the matrix A , containing inductances L_i and mutual inductances M_{ij} , which is analyzed. It can be predefined that the investigated matrix is symmetric and positive definite. In particular the fact that the matrix is positive definite allows to predefined the behavior of several labels due to underlying characteristics of the matrix - which are as follows [7]:

- 1) all eigenvalues (λ_i) of matrix A are real valued (for our case imaginary valued due to $j\omega$) and positive
- 2) eigenvectors corresponding to different eigenvalues are orthogonal ($x_i * x_j = 0$)
- 3) the sum of eigenvalues equals the trace of the matrix, therefore

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n L_i \quad (7)$$

- 4) for $k_{ij}=1$ ($M_{ij} = \sqrt{L_i \cdot L_j}$, see equation (3)) only one eigenvalue exists, all other eigenvalues are zero - therefore the value of this eigenvalue has to be $\sum L_i$, due to statement 3.

Based on the previous description an eigenvalue decomposition is carried out, as this eigenvalues can be seen as (indirect) proportional to the resulting resonance frequency

$$f_{res,i} = \frac{1}{2\pi\sqrt{\lambda_i C}} \quad (8)$$

Note that the capacity of the label is necessary to calculate the resonance frequency. To make the behavior predictable and more obvious, an analysis based on only two labels is carried out first, leading to the following results (figure 4) as function of the coupling coefficient.

It becomes obvious that for a coupling factor of zero no detuning takes place (both labels show the expected resonance frequency of 13.56 MHz) while this detuning becomes more and more significant for a higher coupling factor. The behavior proves finally statement 4 for the case of a coupling factor of 1 - only one eigenvalue exists while the other is zero. In terms of resonance frequency this means that the seen resonance frequency is the lowest achievable one while the other frequency is infinity. The main insight of this discussion is that for two labels there will be two seen resonance frequencies, where the one goes down to lower frequencies with the increasing coupling factor and the other goes up to higher frequencies. An equivalent statement can be formulated for n inlets, resulting in n resonance frequencies. To characterize the behavior of every single frequency would be difficult and is not in the scope of this paper. Nevertheless, the most dominant frequency is the lowest one and therefore it is of essential advantage to know the lowest appearing frequency for a defined number of labels. Here it is in particular

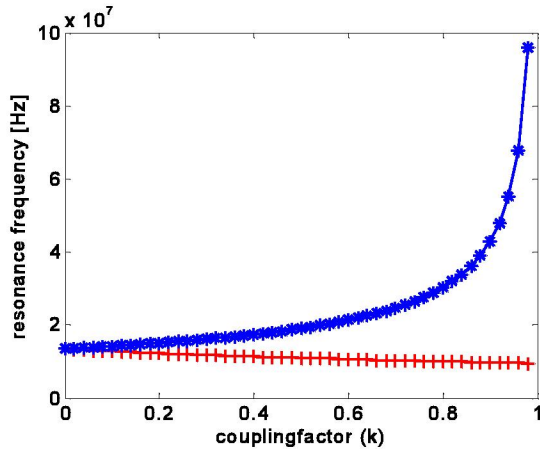


Figure 4: Resonance frequencies based on calculated eigen- values (up to a coupling factor of 0.98).

statement 3 combined with statement 4 which allows to pre-calculate this lowest resonance frequency by

$$f_{res,min} = \frac{1}{2\pi\sqrt{\sum \lambda_i * C}} = \frac{1}{2\pi\sqrt{\sum L_i * C}} \quad (9)$$

Obviously this simple equation allows to predict the value of the most relevant frequency very easily for any number of labels - an essential knowledge when designing a system for applications with several labels in the field.

To prove the above statements, which are based on a pure matrix-analysis, additional Spice Simulations have been carried out including the parameters of the entire label. The simulation is based on the assumption that between driver coil and label 1 is a constant coupling factor while the coupling factor between label 2 and driver coil and label 1 is enhanced continually. The results can be summed up as follows - shown in figure 5:

- Spice Simulations proof the above statements concerning the behavior of the resonance frequency. The investigated A -matrix allows already to predict the principle behavior in terms of resonance frequency absolutely reliable.
- The calculated lower limits are valid also for the entire label.
- The spice based simulation gives further information concerning the behavior of the impedance (amplitude) values, which are shown in figure 5. For the case of close coupled systems it is the lowest appearing frequency which shows the highest amplitude and is therefore the dominant one.
- Finally the diagram points out impressively the change of the quality factor due to the falling resonance frequency. The quality factor again shows a severe influence on the necessary energy to power the IC and/or the reachable distance.

IV. MEASUREMENT

Although the above statements are valuable it is necessary to prove these statements by measurements. The measurements have been carried out by powering the labels through a driver coil of high resonance frequency. Simply by building a stack of labels the resulting lowest resonance frequency has been measured by a Gain-Phase Analyzer, leading to the results shown in figure 6. Do to the stapling the coupling factor shows the highest achievable value (close to one) and therefore allows to compare the measured value with the calculated one. Figure 6 depicts the estimated lowest values based on eigenvalue decomposition additionally, pointing out the excellent correspondence between measurement and calculation. Note that the estimated value due to eigenvalue decomposition is always something lower than the measured value - the reason for this behavior is simply based on the fact that the estimation predicts a coupling factor of 1, which can not be reached by stapling several tags. But especially this represents again the advantage that the estimation points out always the worst case scenario of detuning due to high

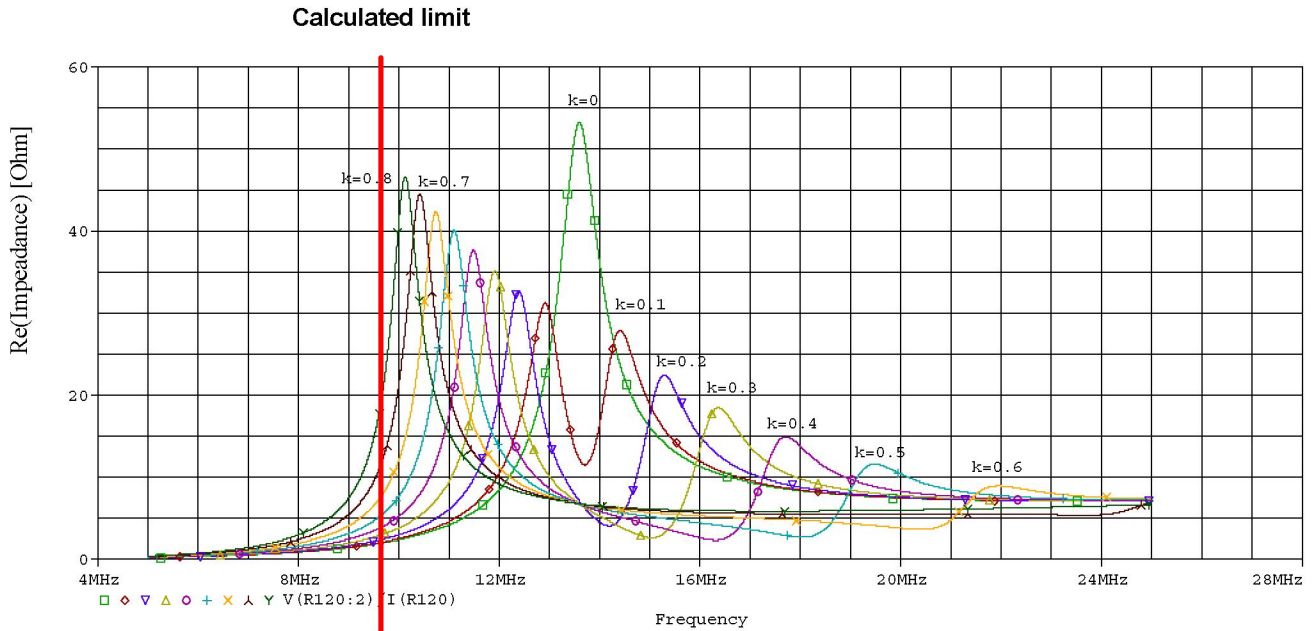


Fig. 5: Spice based simulations

coupled systems and represents therefore a valuable design parameter for future applications.

V. SUMMARY AND OUTLOOK

Topic of this paper was the characterization of multilabel scenarios with a high coupling factor as this scenario will appear in different future applications. It was aim to predict the physical behavior which results due to the detuning effects. In particular the behavior of the resonance frequency has been characterized, as the resonance frequency represents one of the most significant design parameters for inlets. A mathematical method has been developed based on eigenvalue

decom-position which allows to predict the most relevant and dominant resonance frequency very precisely. Further more measurements have shown very impressively that the prediction due to mathematical analysis is correct.

It is topic of further research to characterize the behavior of labels due to a high coupling factor in terms of necessary power, influence on the achievable range etc..

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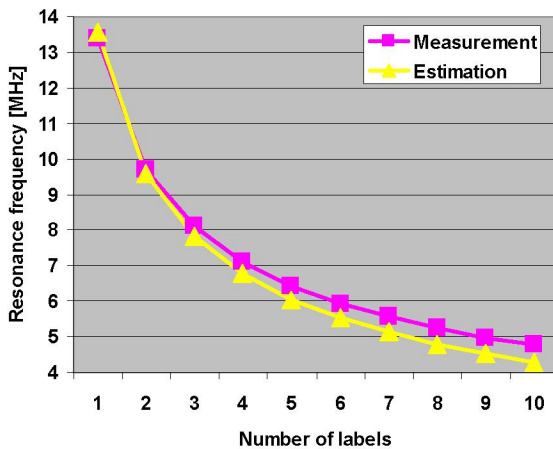


Fig. 6: Measurement versus Estimation